# Lecture 1
# Probability and Bayesian Statistics

From Bayesian statistics to probabilistic machine learning

Probabilistic Machine Learning 2027-1

2026-10-07

Section 1

# A familiar modeling language

# The Beta-Binomial model

**Observed**

the coin tosses $y_1, ..., y_N$

**Unknown**

the coin bias $\theta$

$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$y_i \sim \text{Bernoulli}(\theta), \quad i = 1, ..., N$$

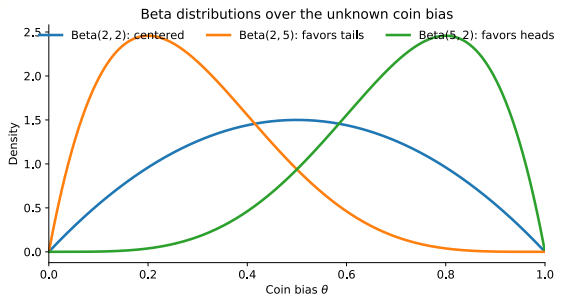
Code heads as $y_i = 1$ and tails as $y_i = 0$

$$N_1 = \sum_{i=1}^{N} y_i, \quad N_0 = N - N_1$$

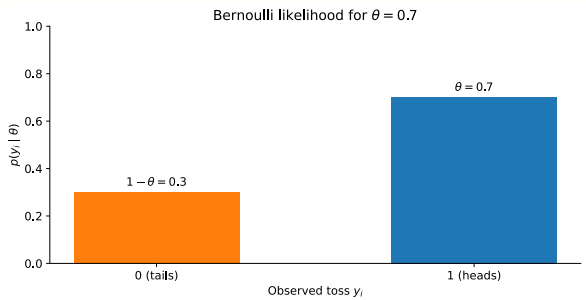
- A statistical model has data and unknown parameters
- How could observed data ($y$) have been generated.
- What does it tell us about the parameters ($\theta$)
- $N_1$ and $N_0$ count heads and tails in the observed sample

# Models are built out of distributions

$$\theta \sim \text{Beta}(\alpha, \beta)$$




$$y_i \sim \text{Bernoulli}(\theta)$$
$$i = 1, \ldots, N$$

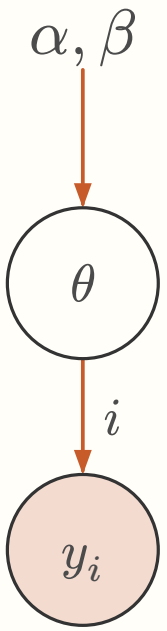



- Beta for continuous variables on $[0, 1]$.
- Bernoulli for each yes/no outcome; Binomial for the count $N_1$.
- Distributions are parameterized families

# Models can be drawn as graphs

$$\theta \sim \text{Beta}(\alpha, \beta)$$

$$y_i \sim \text{Bernoulli}(\theta)$$




- Nodes represent quantities.
  - Shading marks observed data.
- Arrows encode conditional dependence.
  - We will not use Kruschke diagrams

*In this course, graphs will allow us to factor complex problems into smaller, more tractable problems*

# Bayes' rule combines prior and likelihood

**Carried forward** Earlier in this section: Beta–Bernoulli model

Model

$$\theta \sim \text{Beta}(\alpha, \beta)$$
$$y_i \sim \text{Bernoulli}(\theta)$$

**Bayes' rule**

$$p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{p(y)}$$
$$\propto p(y \mid \theta)p(\theta)$$

- Bayes' rule combines prior and likelihood into a posterior
  - **Prior:** $p(\theta)$; **Likelihood:** $p(y \mid \theta)$; **Posterior:** $p(\theta \mid y)$
- We explored the posterior of $\theta$, treating $p(y)$ as a normalizing constant

# Conjugacy permits simple calculations

**Carried forward** Earlier in this section: model and Bayes' rule

Model

$$\theta \sim \text{Beta}(\alpha, \beta)$$
$$y_i \sim \text{Bernoulli}(\theta)$$

Bayes rule

$$p(\theta \mid y) \propto p(y \mid \theta) p(\theta)$$

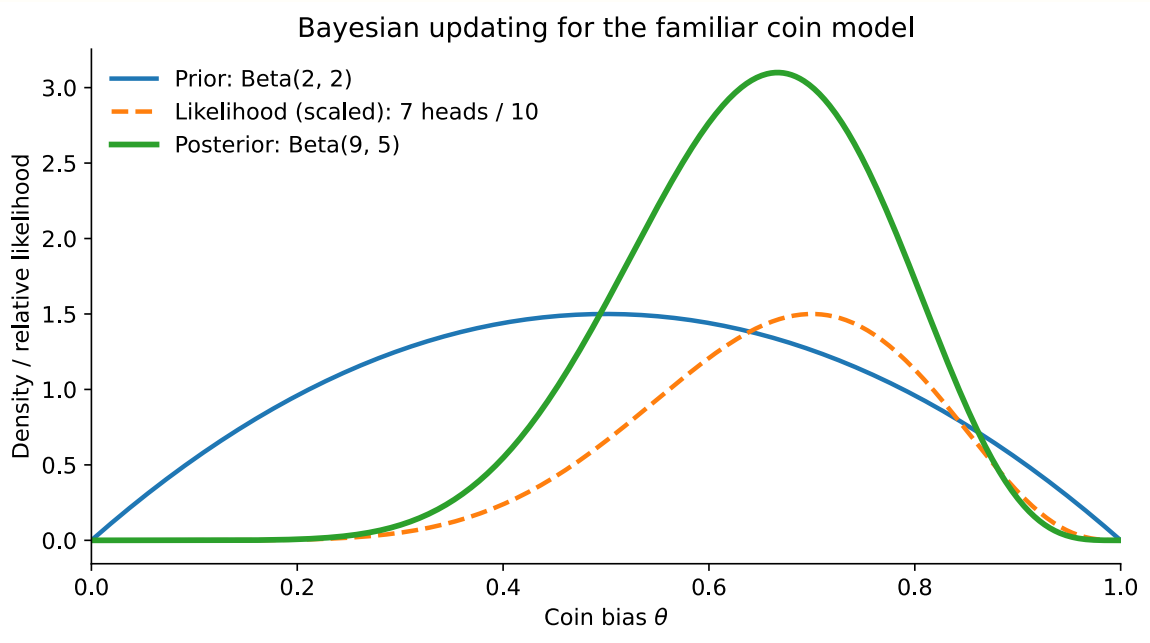
## Applying Bayes' rule to our model

| | |
|---|---|
| **Prior** | $p(\theta) = \text{Beta}(\alpha, \beta)$ |
| **Likelihood** | $p(y \mid \theta) \propto \theta^{N_1}(1-\theta)^{N_0}$ |
| **Posterior** | $p(\theta \mid y) = \text{Beta}(\alpha + N_1, \beta + N_0)$ |

### Conjugacy

The prior and likelihood are conjugate when the posterior belongs to the same distribution family as the prior.

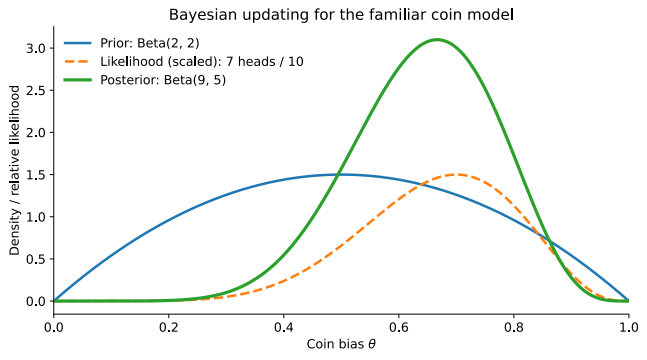
# The posterior is a distribution representing uncertainty




Bayesian updating for the familiar coin model

- The prior and the posterior are both probability distributions
- Posterior width represents remaining uncertainty
- The uncertainty in the posterior is less than in the prior

# Summary: A familiar modeling language




**TAKE HOME**

- A statistical model has data and unknown parameters
- Bayes' rule combines prior and likelihood into a posterior
- Posterior width represents remaining uncertainty

Section 2

# From statistics to machine-learning problems

Same probabilistic language, different organizing questions

# Statistics focuses on the unknown parameters

$$p(\theta \mid D) = \frac{p(D \mid \theta)p(\theta)}{p(D)}$$

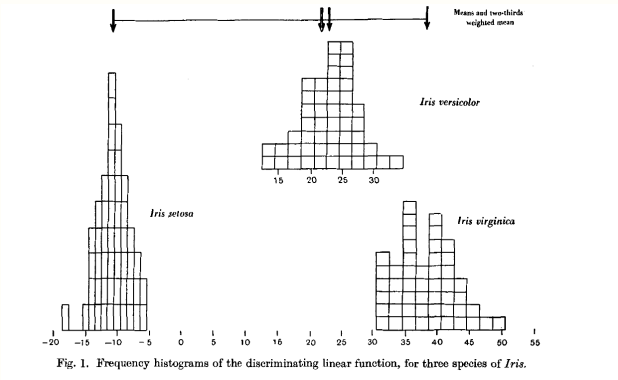
- Get data → build a model → apply Bayes' rule → interpret $\theta$
  - For many scientific questions, that posterior **is** the answer
- Statistics arose from an interest in the underlying parameters in nature

# Machine learning grew out of computer science

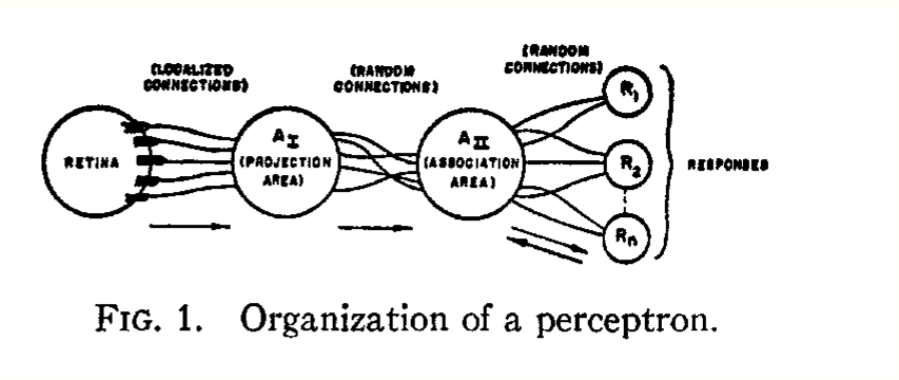



- A computational task is an **input/output contract**
  - Produce a function $f$ that maps input $x$ to output $y$
- Implementation is secondary to the requirement
- Multiple implementations may be possible
- Machine learning developed in this context

# Early machine learning grew out of classification problems



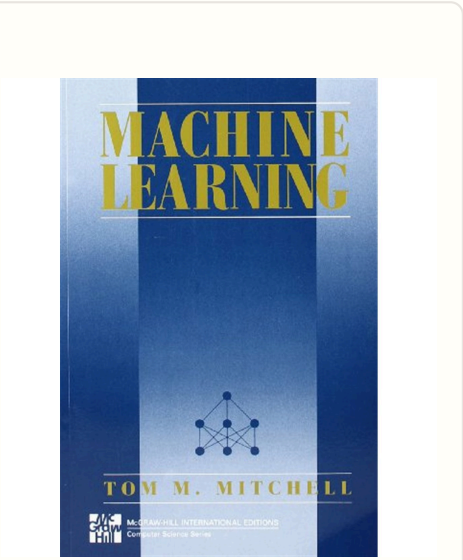

Fisher (1936), Fig. 1, *The Use of Multiple Measurements in Taxonomic Problems*




Rosenblatt (1958), Fig. 1, *The Perceptron*

- Fisher's discriminant and Rosenblatt's perceptron share an interface
  - complex input $\rightarrow$ class or response
- Machine learning asks how data can produce the required function $f$

# Machine learning defined




Tom M. Mitchell, *Machine Learning* (1997), cover

A computer program is said to **learn** from experience $E$ with respect to some class of tasks $T$ and performance measure $P$, if its performance at tasks in $T$, as measured by $P$, improves with experience $E$. ,

| **Training** | **Task** | **Evaluation** |
|---|---|---|
| $f_T : \mathrm{E} \rightarrow f_A$ | $f_A : \mathrm{In} \rightarrow \mathrm{Out}$ | $f_P : (\mathrm{In}, \mathrm{Out}) \rightarrow \mathrm{P}$ |
| $E$: Experience | $f_A$: Task function | $P$: Performance |

- Machine learning is a mapping from $(E, f_P) \rightarrow f_T \rightarrow f_A$
  - That maximizes $f_A$ performance according to $P$

# Supervised experience supports prediction

## Experience: paired examples

$$f_T : \mathrm{E} \quad \to \quad f_A$$

$$f_T : D = \{(x_i, y_i)\}_{i=1}^{N} \quad \to \quad f_D$$

paired examples → learned predictor

## Task: prediction

$$f_A : \mathrm{In} \quad \to \quad \mathrm{Out}$$

$$f_D : x_* \quad \to \quad \hat{y}_*$$

new input → predicted output

- Experience is a training set $D = \{(x_n, y_n)\}$
  - $x$: called feature, covariate, or predictor
  - $y$: called label, target, or response
- Task is to produce a predictor $f_D$ that maps new $x_*$ to correct $y_*$
- Performance is measured by a loss, introduced shortly

# Classification: Iris flowers



Setosa



Versicolor



Virginica

**Table 1.1: subset of the Iris design matrix**

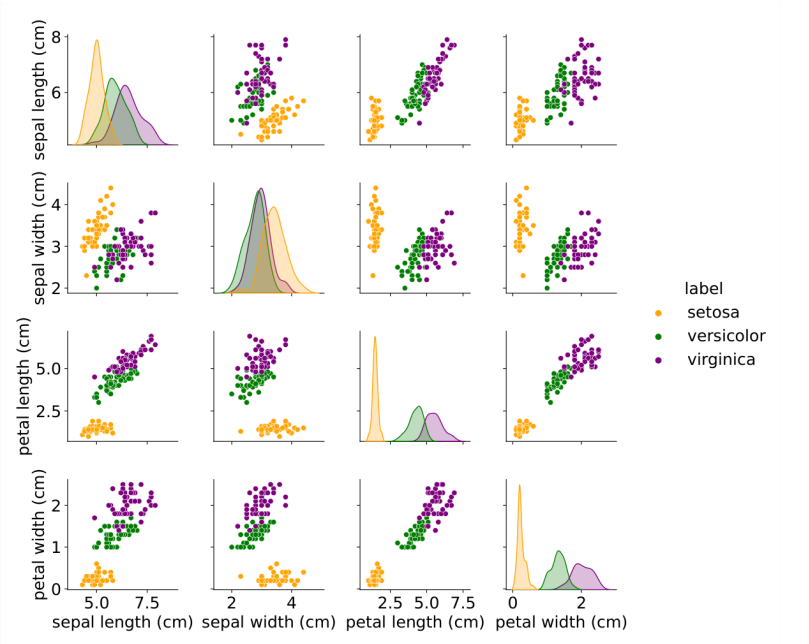
| index | sl | sw | pl | pw | label |
|---|---|---|---|---|---|
| 0 | 5.1 | 3.5 | 1.4 | 0.2 | Setosa |
| 1 | 4.9 | 3.0 | 1.4 | 0.2 | Setosa |
| dots | dots | dots | dots | dots | dots |
| 50 | 7.0 | 3.2 | 4.7 | 1.4 | Versicolor |
| dots | dots | dots | dots | dots | dots |
| 149 | 5.9 | 3.0 | 5.1 | 1.8 | Virginica |

sl: sepal length; sw: sepal width; pl: petal length; pw: petal width

Murphy (2025), Book 1, Fig. 1.1 and Table 1.1

- Classification: $y \in \{1, \ldots, C\}$
  - The classes are unordered and mutually exclusive
- For the Iris dataset
  - $y \in \{\text{Setosa}, \text{Versicolor}, \text{Virginica}\}$
  - $x \in \mathbb{R}^4$

# Exploratory data analysis




Murphy (2025), Book 1, Fig. 1.3; generated by `iris_plot.ipynb`

- Training set
  - 150 labeled examples
  - 50 in each class
- Pair plots shows every pair of features
  - Diagonal shows marginal distributions
- Always inspect data before modeling
  - Structure
  - Outliers
  - Missing values

# Performance: empirical risk minimization

$$L(x, y; \theta) = \ell(y, f(x; \theta))$$

$$L(\theta) = \frac{1}{N} \sum_{n=1}^{N} L(x_n, y_n; \theta)$$

$$\hat{\theta} = \arg\min_{\theta} L(\theta)$$

*Given samples $D = \{x^1, ..., x^N\}$ from $p(X)$, where $X \in \mathbb{R}$, the empirical distribution is $\hat{p}_N(x) = \frac{1}{N} \sum_{n=1}^{N} \delta_{x^n}(x)$.*

- The loss function: the cost of predicting $f(x; \theta)$ when the true label is $y$
- The empirical risk: loss evaluated on the training set
- Empirical risk minimization: minimizing the empirical loss

# A loss function

$$L(x, y; \theta) = \ell(y, f(x; \theta))$$

$$\ell(y, f(x; \theta)) = \ell_{01}(y, \hat{y}) = \mathbb{I}(y \neq \hat{y})$$

$$\mathbb{I}(e) = \begin{cases} 1 \text{ if } e \text{ is true} \\ 0 \text{ if } e \text{ is false} \end{cases}$$

- A loss function assigns a cost to a prediction when the true label is known
- The zero-one loss is one for a misclassification and zero for a correct classification
- $\mathbb{I}(e)$ is an indicator function: it evaluates whether condition $e$ holds

# A probabilistic perspective

$$p(x \mid y = c, \theta)$$

$$p(y = c \mid x, \theta) \propto p(x \mid y = c, \theta) p(y = c)$$

$$\hat{y} = \arg\max_{c \in \mathcal{Y}} p(y = c \mid x; \theta)$$

- The exploratory data analysis suggests:
  - Features in each class have similar distributions
  - Different classes have different distributions
- Bayes' rule converts those ingredients to posterior class probabilities
- A possible classifier: choose the class with maximum a posteriori probability

# Task, experience, and performance

**Task**

What must the learned system do?

$$x \in \mathbb{R}^4 \to \hat{y} \in \mathcal{Y}$$

four measurements → species

**Experience**

What data, feedback, or interaction is available?

$$D = \{(x_n, y_n)\}_{n=1}^{N}$$

150 labeled flowers

**Performance**

How is success evaluated?

$$L(\theta) = \frac{1}{N} \sum_{n=1}^{N} \mathbb{I}(y_n \neq f(x_n; \theta))$$

zero-one loss / misclassification rate

- A learning problem is defined by three ingredients: task, experience, and performance
  - **Task:** predict the Iris species from four sepal and petal measurements
- **Experience:** 150 labeled flowers, with 50 examples from each species
- **Performance:** mean zero-one loss, the training misclassification rate

# Summary: From statistics to machine learning

**Training**

$f_T : \mathrm{E} \to f_A$
$E$: experience

**Task**

$f_A : \mathrm{In} \to \mathrm{Out}$
$f_A$: learned function

**Evaluation**

$f_P : (\mathrm{In}, \mathrm{Out}) \to P$
$P$: performance

**TAKE HOME**

- Machine learning asks how data can produce the required function $f$
- A learning problem is defined by three ingredients: task, experience, and performance

Section 3

# Classes of machine learning problems

Experience regimes, task classes, and performance criteria

# Experience taxonomy

**Supervised**

$$D = \{(x_n, y_n)\}$$

**Unsupervised**

$$D = \{x_n\}$$

**Self-supervised**

$$D = \{x_n, f(x_n)\} \text{ or } \{f(x_n), x_n\}$$

**Reinforcement / interaction**

$$\begin{aligned} \text{decision} &: x(t) \to a(t) \\ \text{action} &: (x(t), a(t)) \to x(t+1) \\ \text{reward} &: x(t+1) \to r(t+1) \end{aligned}$$

> Experience describes what the learner gets to observe

# Task taxonomy

**Prediction**

Task: $\{(x_n, y_n)\} \to \hat{y} = f_D(x_*)$

Performance: $f_P(y_*, \hat{y})$

**Generation**

Task: $\{x_n\} \to (\{c\}, p_D(x \mid c)) \to x_* \sim p_D(x \mid c)$

Performance: $f_P(p(x \mid c), x_*)$

**Discovery**

Task: $\{x_n\} \to (\{z\}, f_D(x)) \to \hat{z} = f_D(x_*)$

Performance: $f_P(\hat{z}, x_*)$

**Action**

Task: $D \to a_* \sim \pi_D(\cdot \mid \{x_{t,*}\}) \to (x_{t+1}, r_{t+1}) = a_t(x_t)$

Performance: $f_P(\{r_t\})$

Task describes what the learned system must do.

| | |
|---|---|
| **Prediction** | Map inputs to outputs |
| **Generation** | Produce new examples |
| **Discovery** | Find a representation of the data |
| **Action** | Choose actions to achieve desired outcomes |

# There are many combinations of experience and task

| Experience | Prediction | Generation | Discovery | Action |
|---|---|---|---|---|
| **Supervised** | Classification | — | — | Imitation learning |
| **Unsupervised** | — | Density modeling | Clustering / PCA | — |
| **Self-supervised** | Next-token prediction | Language generation | Representation learning | — |
| **Reinforcement / interaction** | — | — | — | Policy learning |

- Not every type of experience supports every task.
- Common pairings do exist.

# Summary: Two taxonomies, three ingredients

## Experience: what the learner observes

**Supervised**
$D = \{(x_n, y_n)\}$

**Unsupervised**
$D = \{x_n\}$

**Self-supervised**
$x_n, f(x_n)$

**Interaction**
$x_t \to a_t \to r_{t+1}$

## Task: what the system must do

**Prediction**
$x_* \to \hat{y}_*$

**Generation**
$x_* \sim p_{D(x|c)}$

**Discovery**
$x_* \to \hat{z}_*$

**Action**
$x_t \to a_t$

**TAKE HOME**

- Experience describes what the learner gets to observe
- Task describes what the learned system must do
- Not every type of experience supports every task; common pairings do exist

Section 4

# From datasets to learning problems

The same dataset can pose different learning problems

# Popular datasets

## Iris





AGE
TASKS

Since 1936 – about 90 years
species prediction; clustering; visualization

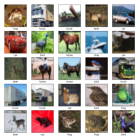
## MNIST



AGE
TASKS

Since 1998 – about 28 years
digit classification; representation learning; generation

## CIFAR-10



AGE
TASKS

Since 2009 – about 17 years
object classification; robustness; generation

## IMDB movie reviews

1. this film was just brilliant casting location scenery story direction everyone's really suited the part they played robert <UNK> is an amazing actor ...
2. big hair big boobs bad music and a giant safety pin these are the words to best describe this terrible movie i love cheesy horror movies and i've seen hundreds...

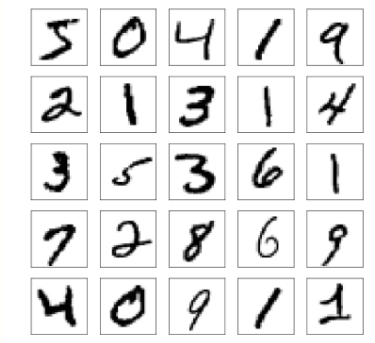
AGE
TASKS

Since 2011 – about 15 years
sentiment classification; text representation

Murphy (2025), Book 1, Figs. 1.1, 1.12a, 1.13b, and Table 1.3

- Machine learning uses common datasets for practice and example problems
- The same dataset can support different learning tasks
- Datasets support benchmarks that make comparisons easy

# MNIST: the same taxonomy, one dataset



$$x_n \in \{0, \ldots, 255\}^{28 \times 28}, \quad y_n \in \{0, \ldots, 9\}$$

Murphy (2025), Book 1, Fig. 1.12a

**Prediction**

$$\hat{y} = f_D(x_*)$$

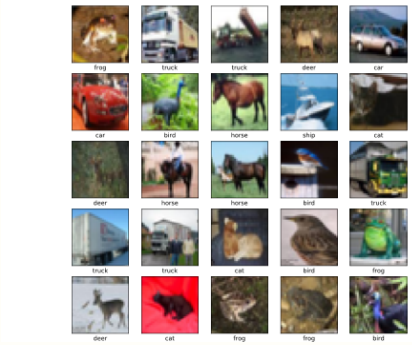
**Discovery**

$$\hat{z} = f_D(x_*)$$

**Generation**

$$x_* \sim p_D(x \mid c)$$

# CIFAR-10: higher-dimensional inputs, the same questions



$$x_n \in \mathbb{R}^{32\times32\times3}, \quad y_n \in \{1, \ldots, 10\}$$

Murphy (2025), Book 1, Fig. 1.13b

**Prediction**

$$\hat{y} = f_D(x_*)$$

$$L(\theta; D_{\text{train}}) \xrightarrow{D_{\text{train}}} L(\theta; D_{\text{test}})$$

# IMDB reviews: text as sequence data

1. this film was just brilliant casting location scenery story direction everyone's really suited the part they played robert <UNK> is an amazing actor ...
2. big hair big boobs bad music and a giant safety pin these are the words to best describe this terrible movie i love cheesy horror movies and i've seen hundreds...

Murphy (2025), Book 1, Table 1.3

**Prediction**

$$\hat{y} = f_D(x_*)$$

**Discovery**

$$\hat{z} = f_D(x_*)$$

**Generation**

$$w_t \sim p_D(w_t \mid w_{<t})$$

# Learning problems become probabilistic queries

**Prediction**

Task: $\{(x_n, y_n)\} \to \hat{y} = f_D(x_*)$

Performance: $f_P(y_*, \hat{y})$

**Generation**

Task: $\{x_n\} \to (\{c\}, p_D(x \mid c)) \to x_* \sim p_D(x \mid c)$

Performance: $f_P(p(x \mid c), x_*)$

**Discovery**

Task: $\{x_n\} \to (\{z\}, f_D(x)) \to \hat{z} = f_D(x_*)$

Performance: $f_P(\hat{z}, x_*)$

**Action**

Task: $D \to a_* \sim \pi_D(\cdot \mid \{x_{t,*}\}) \to (x_{t+1}, r_{t+1}) = a_t(x_t)$

Performance: $f_P(\{r_t\})$

- A learning problem is specified by the variables available and the query we want answered
- Performances are expectations over future cases: $f_P = E[\cdot]$
- Probabilistic notation gives a common language for expressing those queries

# Summary: From datasets to learning problems

## Common practice datasets



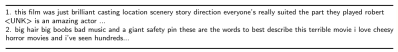
Iris



CIFAR-10

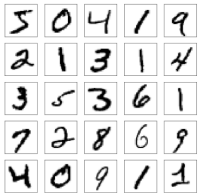


MNIST




IMDB

## One dataset, different tasks



**Prediction**

$\hat{y} = f_{D(x_*)}$

**Discovery**

$\hat{z} = f_{D(x_*)}$

**Generation**

$x_* \sim p_{D(x|c)}$

**TAKE HOME**

- Machine learning uses common datasets for practice and example problems
- The same dataset can support different learning tasks
- A learning problem is specified by the variables available and the query we want answered
- Probabilistic notation gives a common language for expressing those queries

# Summary: Probability for machine learning

Joint distribution: $p_{D(x_{\text{obs}}, z, y_*)}$

**Summarize**
$E[g(X)] = \int g(x)p(x)\,\mathrm{d}x$

**Marginalize**
$p(x) = \int p(x, y)\,\mathrm{d}y$

**Condition**
$p(x|y) = \frac{p(x,y)}{p(y)}$

**TAKE HOME**

- The task determines which quantity we need from the joint distribution
- A small set of recurring operations—summarize, marginalize, condition—answers many of those questions
- Integrals are a central tool for answering these queries

Section 5

# Joint Gaussian distributions and geometry

Describe and transform joint uncertainty with a mean and covariance

# Gaussians: a concrete example

**One variable**

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}}$$

***D* coordinates**

$$\mathcal{N}(\boldsymbol{x} \mid \mu, \Sigma) = \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\mu)^T\Sigma^{-1}(\boldsymbol{x}-\mu)\right)}{(2\pi)^{\frac{D}{2}}\ |\Sigma|^{\frac{1}{2}}}$$

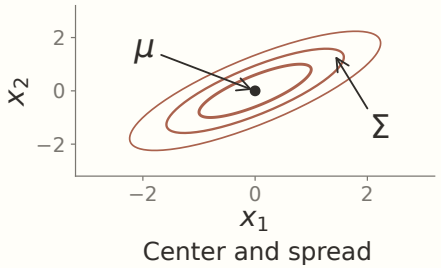
Murphy, Book 2 §2.3.1.1, Eq. 2.71; scalar case: $D = 1$

- One variable: mean $\mu$ and variance $\sigma^2$
- Mean vector $\mu \in \mathbb{R}^D$ locates the cloud; covariance $\Sigma \in \mathbb{R}^{D\times D}$ sets its shape
- Gaussian means, marginals, and conditionals have closed forms

# The three operations in multivariate Gaussian systems

One joint Gaussian: $p(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x} \mid \mu, \Sigma)$

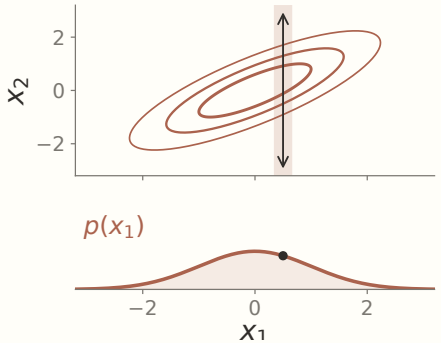
## Summarize




Center and spread

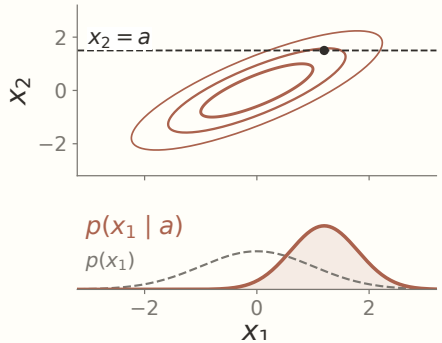
$$\mu = \int \boldsymbol{x} p(\boldsymbol{x}) \, \mathrm{d}\boldsymbol{x}$$

## Marginalize




$$p(x_1) = \int p(x_1, x_2) \, \mathrm{d}x_2$$

## Condition




$$p(x_1 \mid x_2 = a) = \frac{p(x_1, a)}{\int p(u, a) \, \mathrm{d}u}$$

Course figures generated by `gaussian_geometry_figures.ipynb`

- **Summarize:** describe the center and spread with $\mu$ and $\Sigma$
- **Marginalize:** integrate over the coordinate we ignore
- **Condition:** fix an observed coordinate and normalize the slice

# Covariance summarizes joint variation

For a vector $\boldsymbol{x}$ with mean $\mu = E[\boldsymbol{x}]$

$$\Sigma_{ij} = \operatorname{Cov}[X_i, X_j]$$
$$= E\big[(X_i - \mu_i)(X_j - \mu_j)\big]$$

$$\Sigma = E\big[(\boldsymbol{x} - \mu)(\boldsymbol{x} - \mu)^T\big]$$
$$= E\big[\boldsymbol{x}\boldsymbol{x}^T\big] - \mu\mu^T$$

For three coordinates

$$\Sigma = \begin{bmatrix} V[X_1] & \operatorname{Cov}[X_1, X_2] & \operatorname{Cov}[X_1, X_3] \\ \operatorname{Cov}[X_2, X_1] & V[X_2] & \operatorname{Cov}[X_2, X_3] \\ \operatorname{Cov}[X_3, X_1] & \operatorname{Cov}[X_3, X_2] & V[X_3] \end{bmatrix}$$

Murphy, Book 1 §3.1.1, Eqs. 3.1–3.4

- Positive / negative covariance: deviations tend to have the same / opposite signs
- Diagonal entries are variances; off-diagonal entries are covariances
- Zero covariance does not imply independence

# Correlation separates direction from units

$$\rho_{ij} = \frac{\mathrm{Cov}[X_i, X_j]}{\sigma_i \sigma_j}, \quad -1 \leq \rho_{ij} \leq 1$$

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \rho_{12}\sigma_1\sigma_2 & \rho_{13}\sigma_1\sigma_3 \\ \rho_{12}\sigma_1\sigma_2 & \sigma_2^2 & \rho_{23}\sigma_2\sigma_3 \\ \rho_{13}\sigma_1\sigma_3 & \rho_{23}\sigma_2\sigma_3 & \sigma_3^2 \end{bmatrix}$$

Murphy, Book 1 §3.1.2, Eq. 3.7; Book 2 §2.3.1.1, Eq. 2.73

- $\rho_{ij}$ describes the direction and strength of linear association
- $\sigma_1, \sigma_2, \sigma_3$ set the coordinate scales

# Affine maps carry Gaussian means and covariances

$$\boldsymbol{y} = A\boldsymbol{x} + b$$

$$E[\boldsymbol{y}] = A\mu + b$$

$$\mathrm{Cov}[\boldsymbol{y}] = A\Sigma A^T$$

**Covariance proof**

Center: $\boldsymbol{d} = \boldsymbol{x} - \mu$

$$\begin{aligned}
\boldsymbol{y} &= A(\boldsymbol{d} + \mu) + b \\
&= A\boldsymbol{d} + (A\mu + b) \\
\boldsymbol{y} - E[\boldsymbol{y}] &= A\boldsymbol{d} \\
\mathrm{Cov}[\boldsymbol{y}] &= E\big[(A\boldsymbol{d})(A\boldsymbol{d})^T\big] \\
&= AE\big[\boldsymbol{d}\boldsymbol{d}^T\big]A^T \\
&= A\Sigma A^T
\end{aligned}$$

Murphy, Book 1 §3.1.1, Eq. 3.5

- A shift changes the mean but not the covariance
- A linear map changes the scale, orientation, and dimension of variation

# Mahalanobis distance uses the cloud's own scale

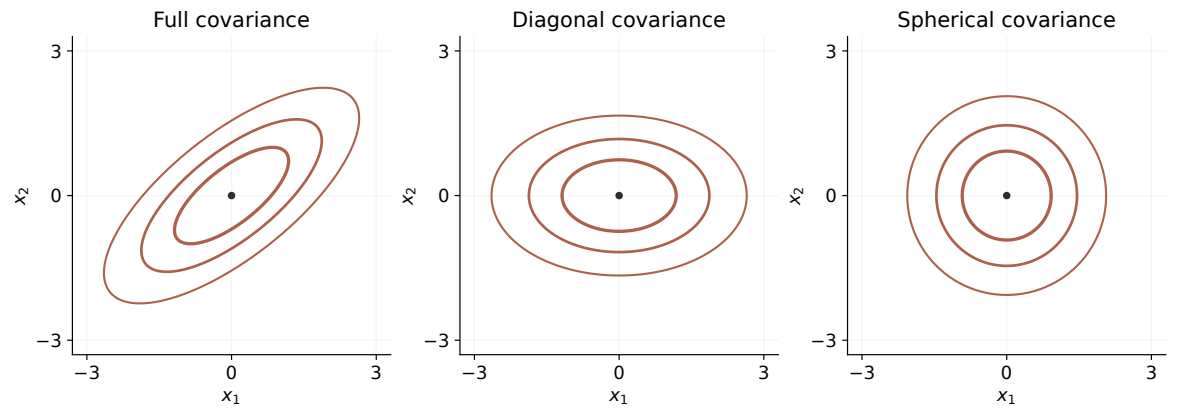
$$d_\Sigma(\boldsymbol{x}, \mu)^2 = (\boldsymbol{x} - \mu)^T \Sigma^{-1} (\boldsymbol{x} - \mu)$$

$$\Sigma = \sigma^2 I_D \quad \rightarrow \quad d_\Sigma^2 = \frac{\|\boldsymbol{x} - \mu\|^2}{\sigma^2}$$

Murphy, Book 2 §2.3.1.1, Eq. 2.72

- Equal Mahalanobis distances trace equal-density contours
- With spherical covariance, it becomes squared Euclidean distance in standard-deviation units

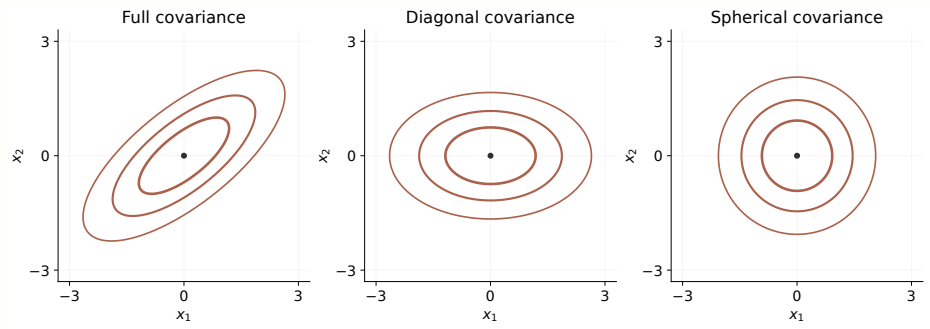
# Covariance controls the contour shape




Course rendering of Murphy, Book 2, Fig. 2.8

- A full covariance can rotate the principal axes
- Diagonal covariance removes linear correlation in these coordinates
- Spherical covariance assigns the same variance in every direction

# Summary: Joint Gaussian geometry




## Affine map

$$\boldsymbol{y} = A\boldsymbol{x} + b$$
$$E[\boldsymbol{y}] = A\mu + b$$
$$\mathrm{Cov}[\boldsymbol{y}] = A\Sigma A^T$$

**TAKE HOME**

- Mean vector $\mu$ locates the cloud; covariance $\Sigma$ sets its shape
- A linear map changes the scale, orientation, and dimension of variation

Section 6

# Marginals and conditionals of a Gaussian

Ignore coordinates or use observations to update uncertainty

# Partition the variables and their parameters together

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}, \quad \Sigma_{21} = \Sigma_{12}^T$$

Murphy, Book 2 §2.3.1.3, Eq. 2.75

- $x_1$ and $x_2$ can each be vectors
- The diagonal blocks describe within-group variation
- The cross blocks describe variation between groups

# Marginalization keeps the matching blocks

$$x \sim \mathcal{N}(\mu, \Sigma)$$

$$p(x_1) = \int p(x_1, x_2)\, \mathrm{d}x_2$$

$$= \mathcal{N}(x_1 \mid \mu_1, \Sigma_{11})$$

$$p(x_2) = \mathcal{N}(x_2 \mid \mu_2, \Sigma_{22})$$

**Marginal proof**

Select $x_1$: $P = [I \;\; 0]$, $\quad x_1 = Px$

By the affine rule:

$$x_1 \sim \mathcal{N}(P\mu, P\Sigma P^T)$$

$$P\mu = \mu_1$$

$$P\Sigma P^T = [I \;\; 0][\Sigma_{11} \;\; \Sigma_{12}]\begin{bmatrix} I \\ 0 \end{bmatrix}$$

$$= \Sigma_{11}$$

Murphy, Book 2 §2.3.1.3, Eqs. 2.76–2.77; affine rule: Book 1 §3.1.1, Eq. 3.5

- Marginalizing $x_2$ integrates over all its possible values
- Cross-covariance affects conditioning, not the marginal of $x_1$

# Conditioning shifts the mean by an observed deviation

$$p(x_1 \mid x_2) = \mathcal{N}\left(x_1 \mid \mu_{1|2}, \Sigma_{1|2}\right)$$

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$

## Mean proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$r = x_1 - \mu_{1|2} = x_1 - \mu_1 - B(x_2 - \mu_2)$$

$$E[r] = 0,$$

$$\mathrm{Cov}[r, x_2] = \Sigma_{12} - B\Sigma_{22}$$

$$= 0$$

$(r, x_2)$ is jointly Gaussian:

zero cross-covariance gives independence

$$E[r \mid x_2] = E[r] = 0$$

$$E[x_1 \mid x_2] = \mu_{1|2}$$

Murphy, Book 2 §2.3.1.3, Eq. 2.78; proof uses affine Gaussian closure

- Conditioning shifts the mean according to the observed deviation $x_2 - \mu_2$, cross-covariance, and observed variance

# Conditioning removes explained covariance

$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

$$\Sigma_{2|1} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$$

## Covariance proof

$$B = \Sigma_{12}\Sigma_{22}^{-1}$$

$$r = x_1 - \mu_{1|2}$$

From the mean proof: $r$ and $x_2$ are independent

Given $x_2$, the mean $\mu_{1|2}$ is constant

$$\Sigma_{1|2} = \mathrm{Cov}[x_1 \mid x_2] = \mathrm{Cov}[r]$$

$$\Sigma_{11} = B\Sigma_{22}B^T + \mathrm{Cov}[r]$$

$$\Sigma_{1|2} = \Sigma_{11} - B\Sigma_{22}B^T$$

$$= \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

Murphy, Book 2 §2.3.1.3, Eqs. 2.78–2.79; proof reuses the independent residual

- Conditioning cannot increase variance in any direction
- In a Gaussian, this covariance depends on which variables are observed, not their values

# The bivariate formula gives intuition

$$x_1, x_2 \in \mathbb{R}$$

$$(X_1 \mid X_2 = x_2) \sim \mathcal{N}\left(\mu_{1|2}, \sigma^2_{1|2}\right)$$

$$\mu_{1|2} = \mu_1 + \rho\frac{\sigma_1}{\sigma_2}(x_2 - \mu_2)$$

$$\sigma_{1|2} = \sigma_1\sqrt{1-\rho^2}$$

Murphy, Book 1 §3.2.4; Book 2 §2.3.1.3

- Positive $\rho$ moves the estimate in the observed direction
- $\rho = 0$ leaves both mean and variance unchanged
- Larger $|\rho|$ leaves less conditional variance

# One observed coordinate can change both the estimate and its uncertainty

$$x_1|2 \sim \mathcal{N}\Big(\mu_1 + \rho\tfrac{\sigma_1}{\sigma_2}(x_2 - \mu_2), (1-\rho^2)\sigma_1^2\Big)$$

$$x_1 \sim \mathcal{N}(0,1), \quad x_2 \sim \mathcal{N}(0,1), \quad \rho = 0.8$$

$$x_2 = 1.5$$

$$x_1 \mid x_2 = 1.5 \sim \mathcal{N}(0.8(1.5), 1 - 0.8^2) = \mathcal{N}(1.2, 0.36)$$

Worked example of Murphy, Book 2 §2.3.1.3, Eq. 2.78

- The observed value lies above its mean, so the conditional mean rises
- The variance falls from 1 to 0.36

# The Schur complement is the residual covariance

**Conditional Gaussian**

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$S = \Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$
$$(x_1 \mid x_2) \sim \mathcal{N}\big(\mu_{1|2}, S\big)$$

**Centered residual**

$$r = x_1 - \mu_{1|2}$$
$$r \sim \mathcal{N}(0, S)$$

The residual $r$ is independent of $x_2$

Remove the part predicted by $x_2$:

$$\mathrm{Cov}\left[\begin{pmatrix} r \\ x_2 \end{pmatrix}\right] = \begin{pmatrix} S & 0 \\ 0 & \Sigma_{22} \end{pmatrix}$$

Murphy, Book 2 §2.3.1.5, Eqs. 2.91, 2.102–2.103

- $S$ measures the uncertainty left after subtracting the conditional mean
- Block elimination leaves $S$: the **Schur complement of $\Sigma_{22}$ in $\Sigma$**

# The joint quadratic separates into conditional and marginal parts

$$p(x_1, x_2) = p(x_1 \mid x_2)p(x_2)$$

$$(\boldsymbol{x} - \mu)^T \Sigma^{-1} (\boldsymbol{x} - \mu)$$

$$= r^T S^{-1} r + (x_2 - \mu_2)^T \Sigma_{22}^{-1} (x_2 - \mu_2)$$

Murphy, Book 2 §2.3.1.5, Eqs. 2.94–2.103

- The first quadratic describes $x_1$ after seeing $x_2$
- The second describes the marginal distribution of $x_2$

# Summary: Marginals and conditionals of a Gaussian

## Marginalize

$$p(x_1) = \int p(x_1, x_2)\, \mathrm{d}x_2$$
$$x_1 \sim \mathcal{N}(\mu_1, \Sigma_{11})$$

## Condition

$$p(x_1|x_2) = \mathcal{N}\left(x_1|\mu_{1|2}, \Sigma_{1|2}\right)$$
$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

**TAKE HOME**

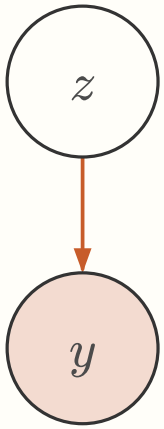
- Marginalizing $x_2$ integrates over all its possible values
- Conditioning shifts the mean according to the observed deviation, cross-covariance, and observed variance
- Conditioning cannot increase variance in any direction

Section 7

# Linear Gaussian systems

Inference from noisy linear measurements

# A linear Gaussian system is a model of state and measurement




Latent state

Observed

## A prior and a measurement law

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)$$

$$y \mid z \sim \mathcal{N}(Wz + b, \Omega)$$

$$p(z, y) = p(z)p(y \mid z)$$

$$y = Wz + b + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \Omega)$$

Noise $\varepsilon$ is independent of the state $z$.

Murphy, Book 2 §2.3.2, Eqs. 2.118–2.119

- This model specifies how an unknown state generates a noisy measurement
- Here the parameters $\mu_z, \Sigma_z, W, b, \Omega$ are given

# One linear Gaussian system supports different questions

$$z \sim \mathcal{N}(\mu_z, \Sigma_z), \quad y \mid z \sim \mathcal{N}(Wz + b, \Omega)$$

**Condition**

$$p(z \mid y) = \frac{p(y \mid z)p(z)}{p(y)}$$

**Summarize**

$$E[g(z) \mid y] = \int g(z)p(z \mid y)\,\mathrm{d}z$$

**Marginalize**

$$p(y) = \int p(y \mid z)p(z)\,\mathrm{d}z$$

Murphy, Book 2 §2.3.2

- Conditioning asks what we can learn from an observed measurement
  - What is the distribution of the hidden state after seeing a measurement?
- Summarizing turns a distribution into a quantity
  - What is our estimate of the hidden state?
- Marginalization averages over an unknown variable to predict another
  - What is the distribution of a measurement before it is observed?

# A linear Gaussian model defines a joint Gaussian

## Linear Gaussian model

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)$$
$$y = Wz + b + \varepsilon$$
$$\varepsilon \sim \mathcal{N}(0, \Omega)$$

$\varepsilon$ is independent of $z$

## Affine Gaussian rules

$$u = Ax + c$$
$$E[u] = AE[x] + c$$
$$\mathrm{Cov}[u] = A\ \mathrm{Cov}[x]A^T$$

## Joint Gaussian

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_z \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{zz} & \Sigma_{zy} \\ \Sigma_{yz} & \Sigma_{yy} \end{pmatrix}\right)$$
$$\mu_y = W\mu_z + b, \quad \Sigma_{zz} = \Sigma_z$$
$$\Sigma_{zy} = \Sigma_z W^T = \Sigma_{yz}^T$$
$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

Murphy, Book 2 §2.3.2.1, Eq. 2.120

- The model plus affine-Gaussian rules determine the full joint distribution
- Once $p(z, y)$ is known, the Gaussian marginal and conditioning rules apply directly

# Bayes' rule for Gaussians

### Model and joint Gaussian

$$\begin{pmatrix} z \\ y \end{pmatrix} \sim \mathcal{N}\left( \begin{pmatrix} \mu_z \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{zz} & \Sigma_{zy} \\ \Sigma_{yz} & \Sigma_{yy} \end{pmatrix} \right)$$

$$\mu_y = W\mu_z + b, \quad \Sigma_{zz} = \Sigma_z$$
$$\Sigma_{zy} = \Sigma_z W^T = \Sigma_{yz}^T$$
$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

### Gaussian conditioning rule

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

### Apply: $x_1 = z, x_2 = y$

$$p(z \mid y) = \mathcal{N}\big(z \mid \mu_{z|y}, \Sigma_{z|y}\big)$$

$$\mu_{z|y} = \mu_z + \Sigma_{zy}\Sigma_{yy}^{-1}\big(y - \mu_y\big)$$

$$\Sigma_{z|y} = \Sigma_{zz} - \Sigma_{zy}\Sigma_{yy}^{-1}\Sigma_{yz}$$

Murphy, Book 2 §2.3.2.2, Eq. 2.121

- Apply the Gaussian conditioning rule to the joint blocks
- The mean responds to observed minus predicted measurement
- The covariance subtracts the uncertainty explained by the measurement

# Bayes' rule as an update formula

**Carried forward** Earlier: Gaussian conditioning

$$\mu_{z|y} = \mu_z + \Sigma_{zy}\Sigma_{yy}^{-1}\left(y - \mu_y\right)$$
$$\mu_y = W\mu_z + b$$
$$\Sigma_{zy} = \Sigma_z W^T$$
$$\Sigma_{yy} = W\Sigma_z W^T + \Omega$$

**Define the gain**

$$K = \Sigma_{zy}\Sigma_{yy}^{-1}$$
$$\mu_{z|y} = \mu_z + K\left(y - \mu_y\right)$$

Murphy, Book 2 §2.3.2.2, Eqs. 2.122–2.124

- $K$ is the gain: it converts a measurement discrepancy into a state correction
- Posterior estimate = prior estimate + gain × prediction error

# The scalar Gaussian update

**Carried forward** Earlier: Gaussian update formula

$$\mu_{z|y} = \mu_z + K(y - \mu_y)$$
$$K = \Sigma_{zy}\Sigma_{yy}^{-1}$$

## Scalar specialization

$$y, z \in \mathbb{R}, \quad W = 1, \quad b = 0$$
$$\Sigma_z = \sigma_z^2, \quad \Omega = \sigma_\varepsilon^2$$
$$z \sim \mathcal{N}(\mu_z, \sigma_z^2)$$
$$y = z + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$$

$\varepsilon$ is independent of $z$

## Predicted mean

$$\mu_y = E[z + \varepsilon] = \mu_z$$

## Cross-covariance

$$\Sigma_{zy} = \mathrm{Cov}[z, z + \varepsilon] = \sigma_z^2$$

## Measurement variance

$$\Sigma_{yy} = \mathrm{Var}[z + \varepsilon] = \sigma_z^2 + \sigma_\varepsilon^2$$

$$K = \frac{\sigma_z^2}{\sigma_z^2 + \sigma_\varepsilon^2} \quad \rightarrow \quad \mu_{z|y} = \mu_z + K(y - \mu_z)$$

Murphy, Book 2 §2.3.2.2; scalar specialization of Eqs. 2.121–2.123

- The scalar gain is prior state variance divided by total measurement variance

# The update is weighted by relative uncertainty

$$\mu_{z|y} = \mu_z + K(y - \mu_z), \quad K = \frac{\sigma_z^2}{\sigma_z^2 + \sigma_\varepsilon^2}$$

◆ = posterior mean $\mu_{z|y}$

**Precise measurement**

$\sigma_\varepsilon^2 \ll \sigma_z^2 \quad K \approx 1$




Trust the measurement

**Equal uncertainty**

$\sigma_\varepsilon^2 = \sigma_z^2 \quad K = \frac{1}{2}$




Split the difference

**Noisy measurement**

$\sigma_\varepsilon^2 \gg \sigma_z^2 \quad K \approx 0$




Trust the prior

Scalar interpretation of Murphy, Book 2 §2.3.2.2

- $K$ is the fraction of the prediction error used in the correction
- More measurement noise gives smaller $K$; more prior uncertainty gives larger $K$

# Inference often combines conditioning and summarizing

**Condition, then summarize**

**Observed** $y$ $\rightarrow$ **Condition** $p(z \mid y) = \mathcal{N}\left(z \mid \mu_{z|y}, \Sigma_{z|y}\right)$ $\rightarrow$ **Summarize** $\hat{z} = E[z \mid y] = \mu_{z|y}$

$$E[z \mid y] = \int z p(z \mid y)\, \mathrm{d}z$$

Murphy, Book 2 §2.3.2.2–2.3.2.3

- Conditioning preserves the whole posterior distribution and its uncertainty
- Summarizing extracts a quantity from that posterior; for a Gaussian the mean is $\mu_{z|y}$

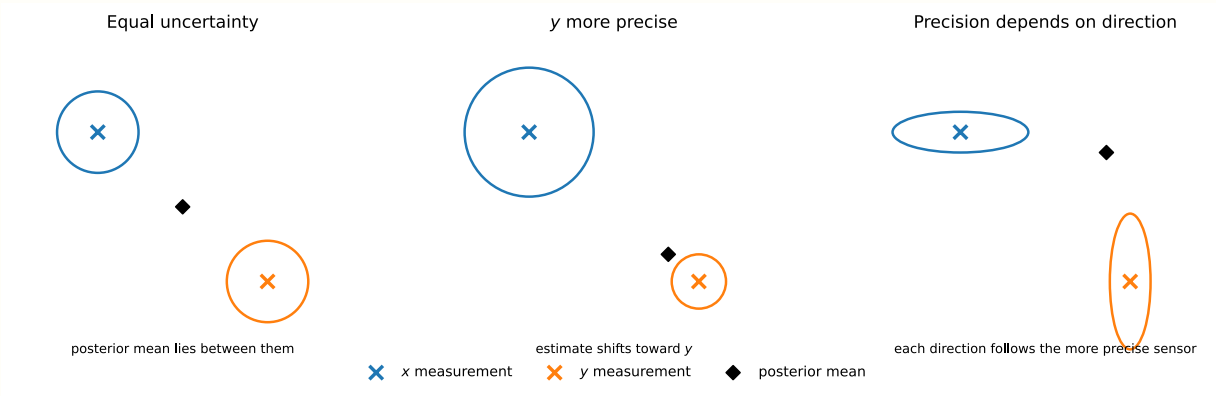
# Sensor fusion combines measurements according to their uncertainty

**Two noisy measurements of one state**

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)\; x \mid z \sim \mathcal{N}(z, \Sigma_x), \quad y \mid z \sim \mathcal{N}(z, \Sigma_y)$$

**Same two operations**

$$p(z \mid x, y) \quad \rightarrow \quad E[z \mid x, y]$$




Course schematic after Murphy, Book 2 §2.3.2.3, Fig. 2.10; `sensor_fusion_figure.ipynb`

- Equal uncertainty puts the posterior mean between the measurements
- A more precise sensor pulls the posterior mean closer to its measurement
- With anisotropic uncertainty, the weighting can differ by direction

# The same model can also predict measurements

**Model and joint**

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)$$
$$y \mid z \sim \mathcal{N}(Wz + b, \Omega)$$
$$p(z, y) = p(z)p(y \mid z)$$

**Marginalization rule**

$$p(y) = \int p(z, y)\, \mathrm{d}z$$
$$= \int p(y \mid z)p(z)\, \mathrm{d}z$$

$$p(y) = \mathcal{N}\left(y \mid \mu_y, \Sigma_{yy}\right) \mu_y = W\mu_z + b, \quad \Sigma_{yy} = W\Sigma_z W^T + \Omega$$

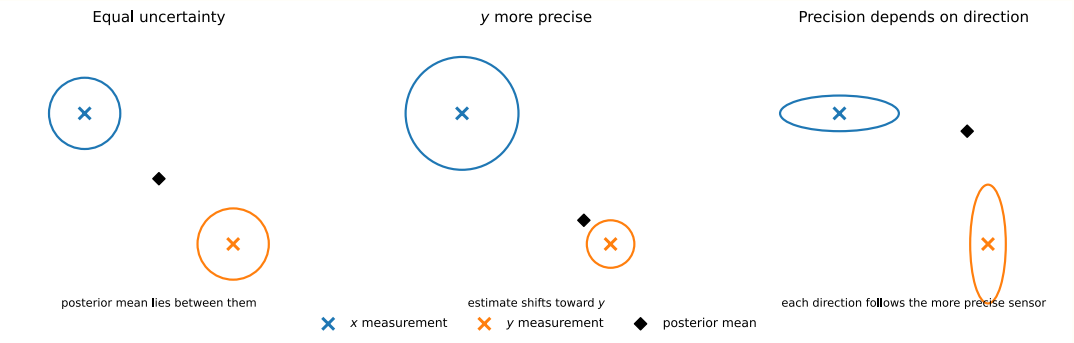
Murphy, Book 2 §2.3.2.2, Eq. 2.129

- Before observing $y$, marginalization gives the distribution of possible measurements
- These are the same $\mu_y$ and $\Sigma_{yy}$ that appear inside the conditioning update

# Summary: Linear Gaussian systems

## Linear Gaussian model

$$z \sim \mathcal{N}(\mu_z, \Sigma_z)$$
$$y \mid z \sim \mathcal{N}(Wz + b, \Omega)$$
$$p(z, y) = p(z)p(y|z)$$




**TAKE HOME**

- The model plus affine-Gaussian rules determine the full joint distribution
- Posterior estimate = prior estimate + gain × prediction error
- More measurement noise gives smaller $K$; more prior uncertainty gives larger $K$

# Lecture 1: four visual anchors

## One dataset, several questions

**A dataset is not a task.**



$x \to \hat{y}$ Prediction
$x \to \hat{z}$ Discovery
$x_* \sim p(x \mid c)$ Generation

## Three probability operations

**The task determines the operation.**

**Summarize** $E[g(X)] = \int g(x)p(x)\,\mathrm{d}x$
**Marginalize** $p(x) = \int p(x, y)\,\mathrm{d}y$
**Condition** $p(x \mid y) = \frac{p(x,y)}{p(y)}$

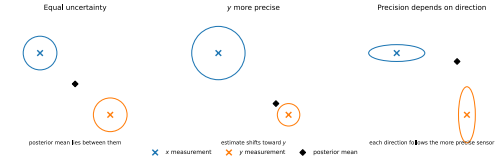
## Gaussian conditioning

**Observation changes estimate and uncertainty.**

$$\mu_{1|2} = \mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2)$$
$$\Sigma_{1|2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$$

## Sensor fusion

**Uncertainty controls how evidence is combined.**




Visual anchors from Lecture 1: MNIST tasks, three probability operations, Gaussian conditioning, and sensor fusion